PPT TERRAFORM

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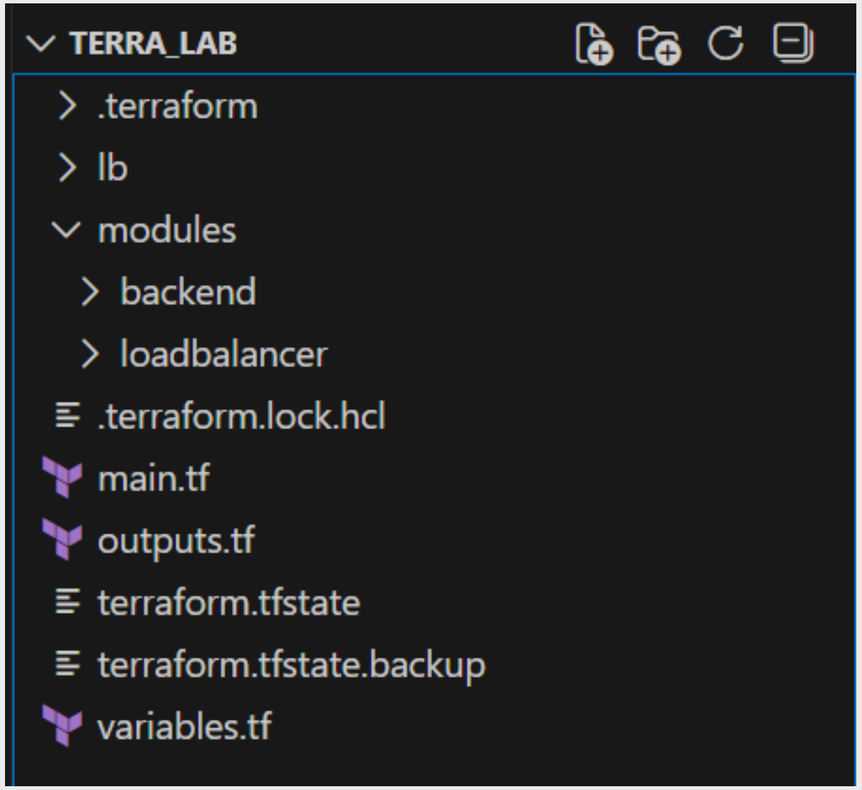
struktur folder



mkdir terra_lab

```
nestt@DESKTOP-AFG98AC:~$ mkdir terra_lab
nestt@DESKTOP-AFG98AC:~$ cd terra_lab
nestt@DESKTOP-AFG98AC:~/terra_lab$ code .
```

terraform init

```
nestt@DESKTOP-AFG98AC:~/terra_lab$ terraform init

Initializing the backend...
Initializing modules...
- backend in modules/backend
- loadbalancer in modules/loadbalancer

Initializing provider plugins...
- Finding kreuzwerker/docker versions matching "~> 3.0"...
- Installing kreuzwerker/docker v3.6.2...
- Installed kreuzwerker/docker v3.6.2 (self-signed, key ID BD080C4571C6104C)

Partner and community providers are signed by their developers.
If you'd like to know more about provider signing, you can read about it here:
https://www.terraform.io/docs/cli/plugins/signing.html

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

terraform plan

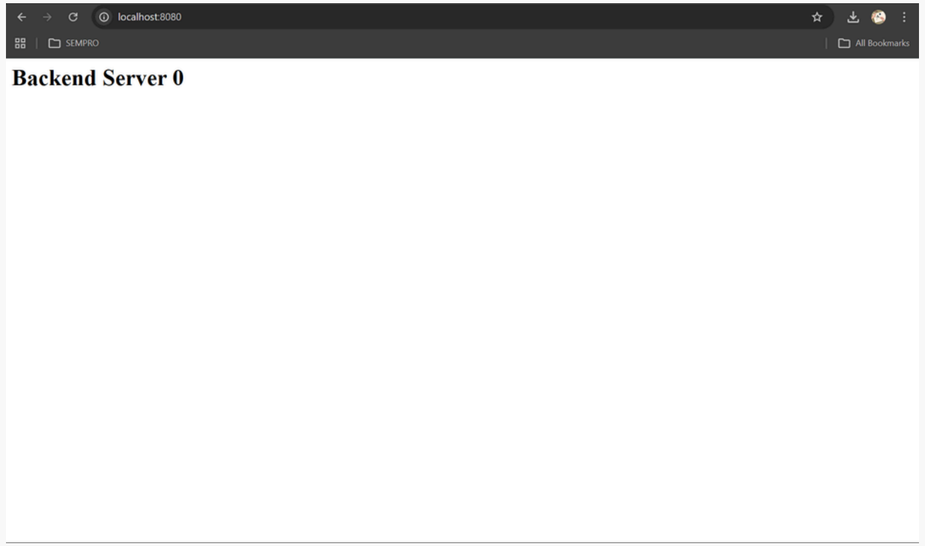
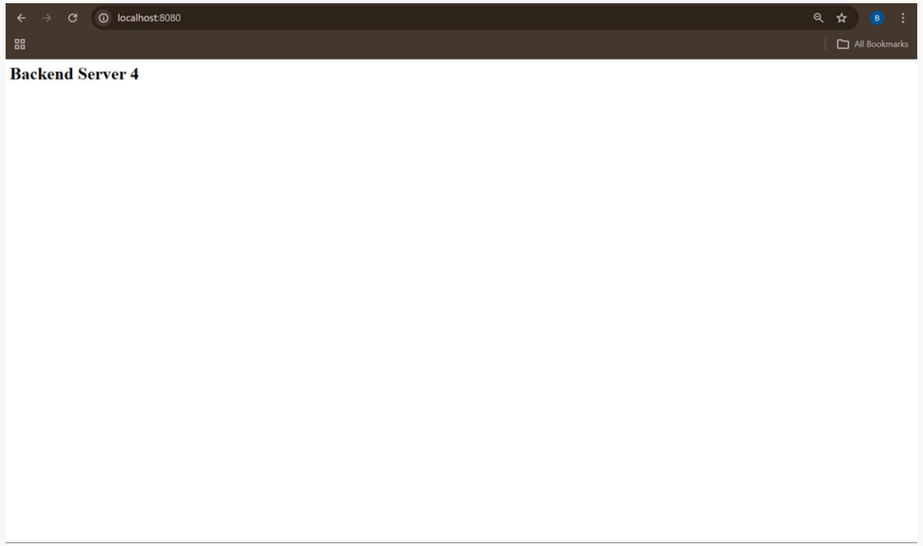
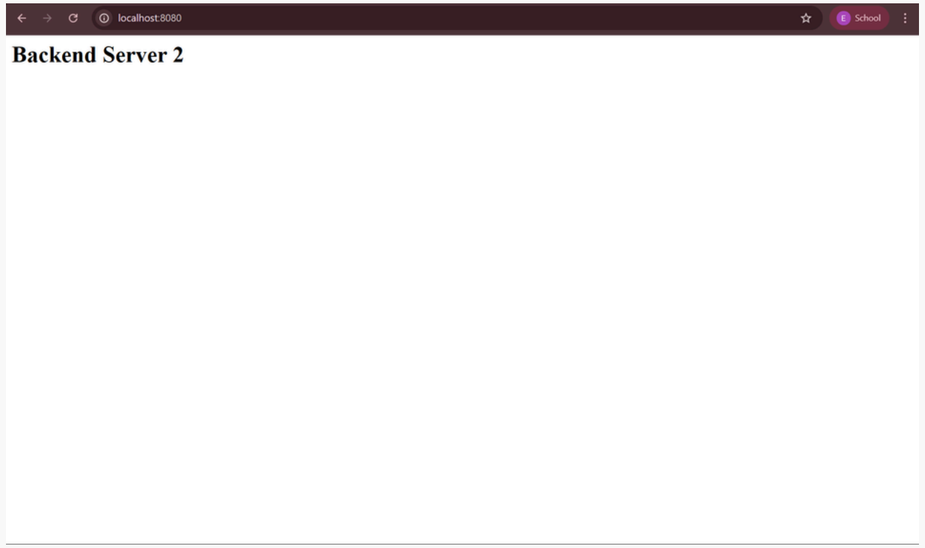
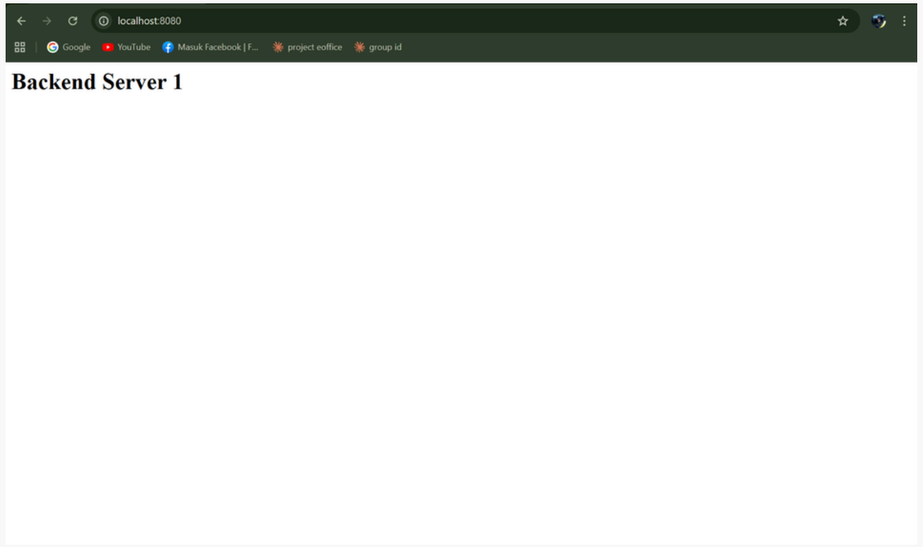
```
nestt@DESKTOP-AFG98AC:~/terra_lab$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  + create

Terraform will perform the following actions:

  # docker_network.app_network will be created
  + resource "docker_network" "app_network" {
      + driver      = (known after apply)
      + id          = (known after apply)
      + internal    = (known after apply)
      + ipam_driver = "default"
      + name        = "mini_network"
      + options     = (known after apply)
      + scope       = (known after apply)
    }

  # module.backend.docker_container.backend[0] will be created
  + resource "docker_container" "backend" {
      + attach                                      = false
      + bridge                                      = (known after apply)
      + command                                     = [
          + "sh",
          + "-c",
          + "echo '<h1>Backend Server 0</h1>' > /usr/share/nginx/html/index.html && nginx -g 'daemon off;'",
        ]
      + container_logs                              = (known after apply)
      + container_read_refresh_timeout_milliseconds = 15000
      + entrypoint                                  = (known after apply)
      + env                                         = (known after apply)
      + exit_code                                   = (known after apply)
      + hostname                                    = (known after apply)
      + id                                          = (known after apply)
      + image                                       = (known after apply)
      + init                                        = (known after apply)
      + ipc_mode                                    = (known after apply)
      + log_driver                                  = (known after apply)
      + logs                                        = false
      + must_run                                    = true
      + name                                        = "backend_0"
      + network_data                                = (known after apply)
      + network_mode                                = "bridge"
      + read_only                                   = false
      + remove_volumes                              = true
      + restart                                     = "no"
      + rm                                          = false
```

```
  }

# module.backend.docker_container.backend[1] will be created
+ resource "docker_container" "backend" {
    + attach                                      = false
    + bridge                                      = (known after apply)
    + command                                     = [
        + "sh",
        + "-c",
        + "echo '<h1>Backend Server 1</h1>' > /usr/share/nginx/html/index.html && nginx -g 'daemon off;'",
      ]
    + container_logs                              = (known after apply)
    + container_read_refresh_timeout_milliseconds = 15000
    + entrypoint                                  = (known after apply)
    + env                                         = (known after apply)
    + exit_code                                   = (known after apply)
    + hostname                                    = (known after apply)
    + id                                          = (known after apply)
    + image                                       = (known after apply)
    + init                                        = (known after apply)
    + ipc_mode                                    = (known after apply)
    + log_driver                                  = (known after apply)
    + logs                                        = false
    + must_run                                    = true
    + name                                        = "backend_1"
    + network_data                                = (known after apply)
    + network_mode                                = "bridge"
    + read_only                                   = false
    + remove_volumes                              = true
    + restart                                     = "no"
    + rm                                          = false
    + runtime                                     = (known after apply)
    + security_opts                               = (known after apply)
    + shm_size                                    = (known after apply)
    + start                                       = true
    + stdin_open                                  = false
    + stop_signal                                 = (known after apply)
    + stop_timeout                                = (known after apply)
    + tty                                         = false
    + wait                                        = false
    + wait_timeout                                = 60

    + networks_advanced {
        + aliases = []
        + name    = "mini_network"
      }
  }
```

```
# module.backend.docker_container.backend[2] will be created
+ resource "docker_container" "backend" {
    + attach                                      = false
    + bridge                                      = (known after apply)
    + command                                     = [
        + "sh",
        + "-c",
        + "echo '<h1>Backend Server 2</h1>' > /usr/share/nginx/html/index.html && nginx -g 'daemon off;'",
      ]
    + container_logs                              = (known after apply)
    + container_read_refresh_timeout_milliseconds = 15000
    + entrypoint                                  = (known after apply)
    + env                                         = (known after apply)
    + exit_code                                   = (known after apply)
    + hostname                                    = (known after apply)
    + id                                          = (known after apply)
    + image                                       = (known after apply)
    + init                                        = (known after apply)
    + ipc_mode                                    = (known after apply)
    + log_driver                                  = (known after apply)
    + logs                                        = false
    + must_run                                    = true
    + name                                        = "backend_2"
    + network_data                                = (known after apply)
    + network_mode                                = "bridge"
    + read_only                                   = false
    + remove_volumes                              = true
    + restart                                     = "no"
    + rm                                          = false
    + runtime                                     = (known after apply)
    + security_opts                               = (known after apply)
    + shm_size                                    = (known after apply)
    + start                                       = true
    + stdin_open                                  = false
    + stop_signal                                 = (known after apply)
    + stop_timeout                                = (known after apply)
    + tty                                         = false
    + wait                                        = false
    + wait_timeout                                = 60

    + networks_advanced {
        + aliases = []
        + name    = "mini_network"
      }
  }

# module.backend.docker_container.backend[3] will be created
```

```
# module.backend.docker_container.backend[2] will be created
+ resource "docker_container" "backend" {
    + attach                                      = false
    + bridge                                      = (known after apply)
    + command                                     = [
        + "sh",
        + "-c",
        + "echo '<h1>Backend Server 2</h1>' > /usr/share/nginx/html/index.html && nginx -g 'daemon off;'",
      ]
    + container_logs                              = (known after apply)
    + container_read_refresh_timeout_milliseconds = 15000
    + entrypoint                                  = (known after apply)
    + env                                         = (known after apply)
    + exit_code                                   = (known after apply)
    + hostname                                    = (known after apply)
    + id                                          = (known after apply)
    + image                                       = (known after apply)
    + init                                        = (known after apply)
    + ipc_mode                                    = (known after apply)
    + log_driver                                  = (known after apply)
    + logs                                        = false
    + must_run                                    = true
    + name                                        = "backend_2"
    + network_data                                = (known after apply)
    + network_mode                                = "bridge"
    + read_only                                   = false
    + remove_volumes                              = true
    + restart                                     = "no"
    + rm                                          = false
    + runtime                                     = (known after apply)
    + security_opts                               = (known after apply)
    + shm_size                                    = (known after apply)
    + start                                       = true
    + stdin_open                                  = false
    + stop_signal                                 = (known after apply)
    + stop_timeout                                = (known after apply)
    + tty                                         = false
    + wait                                        = false
    + wait_timeout                                = 60

    + networks_advanced {
        + aliases = []
        + name    = "mini_network"
      }
  }

# module.backend.docker_container.backend[3] will be created
```

terraform apply

```
      + volumes {
          + container_path = "/etc/nginx/nginx.conf"
          + host_path      = "/home/nestt/terra_lab/lb/nginx.conf"
        }
    }

  # module.loadbalancer.docker_image.nginx will be created
  + resource "docker_image" "nginx" {
      + id          = (known after apply)
      + image_id    = (known after apply)
      + name        = "nginx:latest"
      + repo_digest = (known after apply)
    }

Plan: 9 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + backend_container_names = [
      + "backend_0",
      + "backend_1",
      + "backend_2",
      + "backend_3",
      + "backend_4",
    ]
  + load_balancer_url       = "http://localhost:8080"

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

module.backend.docker_image.nginx: Creating...
docker_network.app_network: Creating...
module.loadbalancer.docker_image.nginx: Creating...
docker_network.app_network: Creation complete after 3s [id=54befd663699d91073758d425148ddccf25396b43d524fc9aa47af4e45b8cb91]
module.loadbalancer.docker_image.nginx: Still creating... [10s elapsed]
module.backend.docker_image.nginx: Still creating... [10s elapsed]
module.backend.docker_image.nginx: Creation complete after 20s [id=sha256:fd204fe2f75024354b1f979d38cc43def9e049cc2df1cda45074d1b84c4f9b3enginx:latest]
module.loadbalancer.docker_image.nginx: Creation complete after 20s [id=sha256:fd204fe2f75024354b1f979d38cc43def9e049cc2df1cda45074d1b84c4f9b3enginx:latest]
module.backend.docker_container.backend[3]: Creating...
module.backend.docker_container.backend[1]: Creating...
module.backend.docker_container.backend[2]: Creating...
module.backend.docker_container.backend[4]: Creating...
module.backend.docker_container.backend[0]: Creating...
module.loadbalancer.docker_container.loadbalancer: Creating...
module.backend.docker_container.backend[2]: Creation complete after 1s [id=bab5a739e3af51881ac92af2025e9e78e15ed3f259bd2dbce6c30dd784effab4]
module.backend.docker_container.backend[3]: Creation complete after 1s [id=2382beb1808341b47e6f9d676c04ade5a5251e903ceb6d131d6f25ff79cde2b6]
module.backend.docker_container.backend[4]: Creation complete after 1s [id=372ea92078e846dde5443f305819a7c1db22c6a0d1d919628e46ec04e3b2f5b9]
module.backend.docker_container.backend[0]: Creation complete after 1s [id=41491026b703ef0931c50901c6682b895df436fa233d1ed0d58999b5b48e8220]
module.backend.docker_container.backend[1]: Creation complete after 1s [id=4ba95cd63f219afae07190b8dd27f180de49048ed7a0ec16dda48d9b34299df7]
module.loadbalancer.docker_container.loadbalancer: Creation complete after 1s [id=d34257cc52c2c4e0c73546aa0c08e871e2319574b47f2ac4566108e4baf4cbba]

Apply complete! Resources: 9 added, 0 changed, 0 destroyed.

Outputs:
```

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